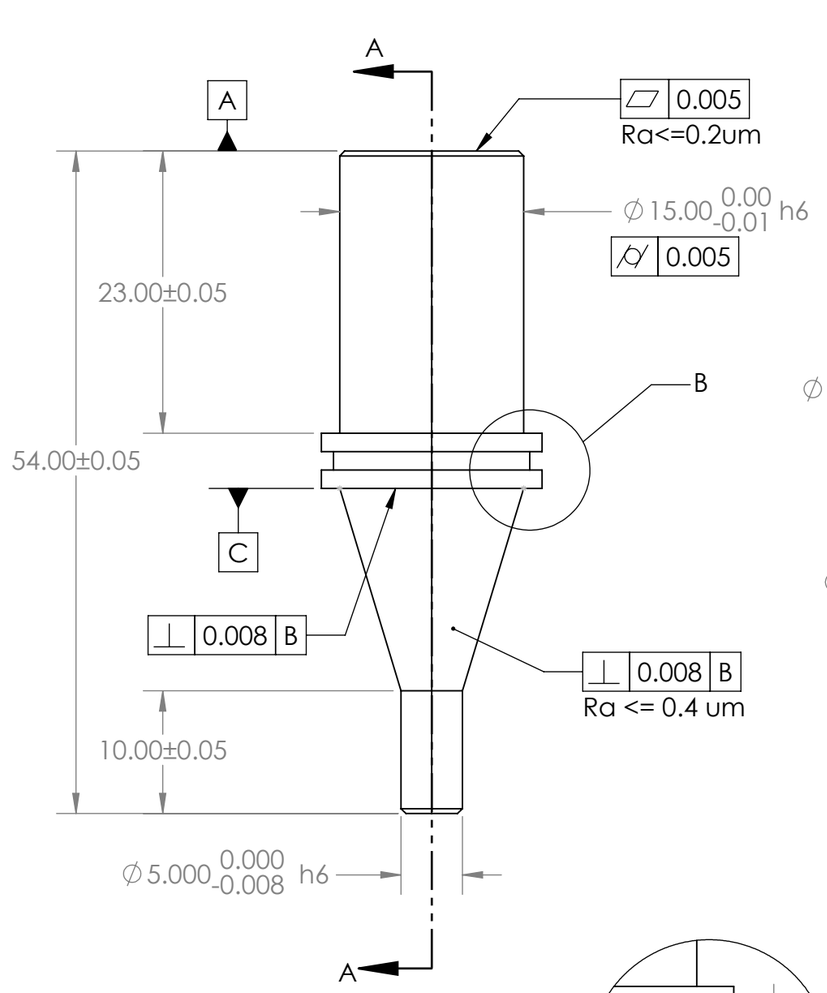
GENERAL NOTES: UNLESS OTHERWISE SPECIFIED

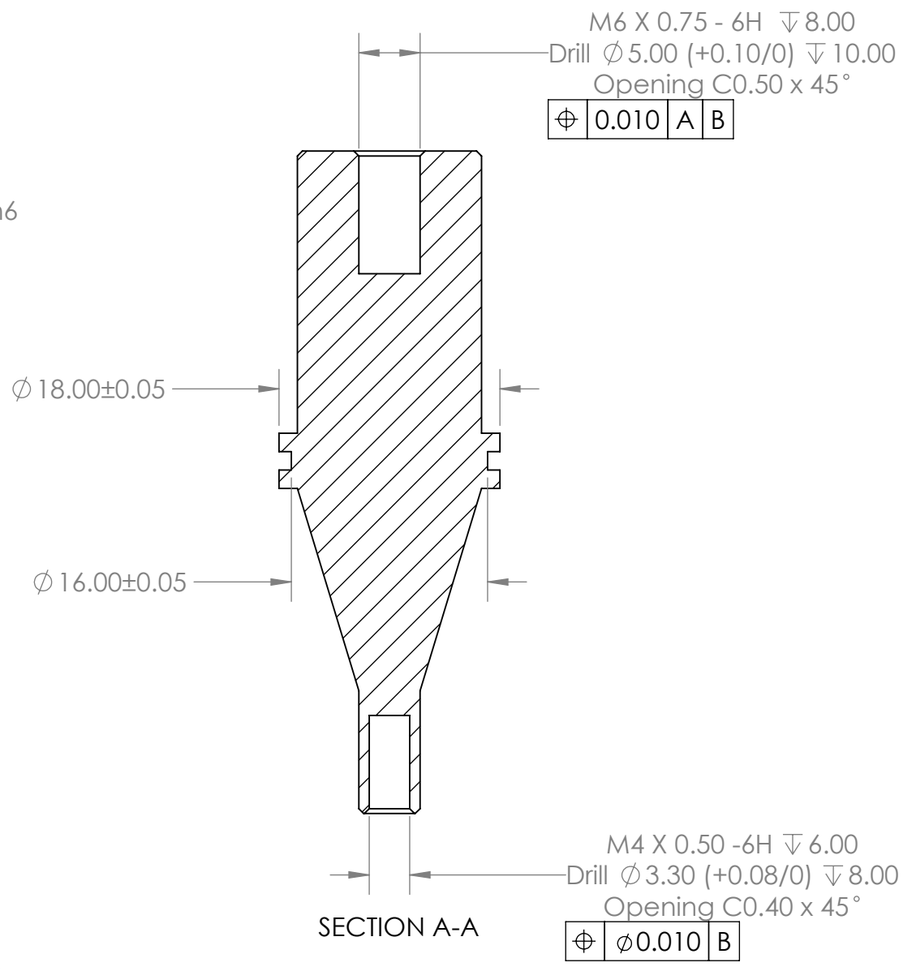
1. GENERAL & STANDARDS COMPLIANCE:
1.1 INTERPRET DRAWING IN ACCORDANCE WITH ASME Y14.5-2018 AND ISO 8015.
1.2 3D CAD DATASET (W3-MED-US-HORN-5.4.STEP) GOVERNS ACOUSTIC CONTOUR FOR UNTOLERANCED SURFACES.
1.3 GENERAL TOLERANCES: LINEAR +/-0.05 mm, ANGULAR +/-0.1 DEG (ULTRA-PRECISION MEDICAL ACOUSTIC MACHINING).
1.4 THIRD ANGLE PROJECTION APPLIES THROUGHOUT. ALL UNITS IN MILLIMETERS.

2. MATERIAL & ACOUSTIC PERFORMANCE SPECIFICATIONS:
2.1 MATERIAL: MEDICAL GRADE TITANIUM ALLOY TC4 ELI (Ti-6Al-4V GRADE 23 / ASTM F136 / ISO 5832-3), EXTRA LOW INTERSTITIAL.
2.2 MECHANICAL PROPERTIES: YIELD STRENGTH Rp0.2 >= 790 MPa, TENSILE STRENGTH Rm >= 860 MPa, ELONGATION A >= 10%. ACOUSTIC SPEED c >= 5050 m/s.
2.3 100% HOMOGENEOUS SUBTRACTIVE MACHINING ONLY (PRECISION CNC TURN-MILL). ZERO RESIDUAL MICRO-POROSITY OR INCLUSIONS.

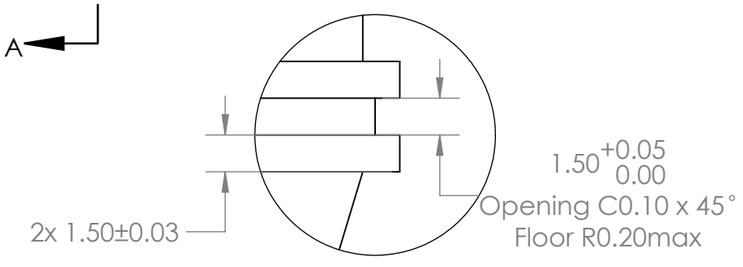
3. ACOUSTIC DYNAMICS & GD&T CONTROLS:
3.1 PRIMARY DATUM [A] (PZT STACK SEAT): FLATNESS <= 0.005 mm, SURFACE FINISH Ra <= 0.2 MICRONS. ZERO SCRATCHES, COAXIAL RUNOUT RELATIVE TO [B] WITHIN 0.008 mm.
3.2 SECONDARY DATUM [B] (LONGITUDINAL CYLINDRICAL AXIS): TOTAL RADIAL RUNOUT ACROSS ENTIRE PROFILE WITHIN 0.010 mm RELATIVE TO [B].
3.3 THREAD COAXIALITY: M6x0.75-6H AND M4x0.5-6H THREADS MUST BE COAXIAL TO DATUM [B] WITHIN Ø0.010 mm TO PREVENT PARASITIC TRANSVERSE VIBRATION MODES.
3.4 RESONANCE SPECIFICATION: DESIGNED FOR 55.5 kHz (+/- 500 Hz) LONGITUDINAL HALF-WAVELENGTH RESONANCE MODE. AMPLIFICATION RATIO M >= 3.0.

4. SURFACE FINISH & FATIGUE MITIGATION:
4.1 CRITICAL TRANSITION PROFILE FINISH: Ra <= 0.4 MICRONS. ALL CUTTER MARKS MUST BE LONGITUDINAL OR REMOVED VIA AXIAL MICRO-POLISHING TO ELIMINATE NOTCH-SENSITIVE FATIGUE NUCLEATION SITES.
4.2 NODAL FLANGE RETENTION LIPS: MINIMUM SOLID RADIAL/AXIAL LIP THICKNESS MUST BE >= 1.50 mm (100% DEFECT-FREE, NO RESIDUAL BURRS).





SECTION A-A



DETAIL B

UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN MILLIMETERS SURFACE FINISH: TOLERANCES: Linear ±0.05 mm, Angular ±0.1° (Ultra-Precision Lathe / Turn-Mill)	FINISH: Acoustic Mirror Polish (Ra <= 0.4 µm along profile; Ra <= 0.2 µm on PZT datum face)				DEBURR AND BREAK SHARP EDGES	DO NOT SCALE DRAWING	REVISION: A
	NAME	SIGNATURE	DATE			TITLE: 55.5 kHz ULTRASONIC SURGICAL SCALPEL ACOUSTIC HORN	
DRAWN							
CHK'D							
APPV'D							
MFG							
Q.A				MATERIAL: Medical Grade Titanium Alloy TC4 ELI (Ti-6Al-4V Grade 23 / ASTM F136 / ISO 5832-3)		DWG NO. W3-MED-US-HORN-5.4	A3
				WEIGHT: ~28.64 g		SCALE : NTS	SHEET : 1 OF 1